Measure before you improve

Day 07: 81 labelled questions, two retrievers, every miss explained.

31 QUESTIONS: IS THE ANSWER IN THE TOP 3?



same total, different misses

1. lost laptop: only BM25 finds it
2. new dads: only dense finds it
3. wine (alcohol): both miss it

30 of 31 found by at least one: combine them? → Day 08

30 LOOKUPS: A VALUE TYPED AS THE QUERY

USD 220, 15 minutes...



15 MISSES, EACH WITH A WRITTEN REASON

11	exact value	dense	2	vocabulary	BM25
1	other section	dense	1	broader chunk	dense

each points at a fix a later day will test, on these same questions

A label is evidence

The right answer depends on the documents, and on who asks.

"When do pay rises take effect?"

evidence `take effect from 1 April`

only in `manager-pay-review-guide#overview` — managers only

maria (manager)
label: that chunk
answerable

alice (employee)
label: none she may read
correct: "I don't know"

LABELS BELONG TO A SNAPSHOT — 44 chunks: text hash + who may read

password rule: 14 → 16 characters — 2 labels break

retention policy → managers only — 4 of alice's labels flip

17 label problems → the notebook stops before measuring

DEV · 54

Days 01–06 questions
seen while building:
tune on these
16 answerable · 8 not · 30 lookups

TEST · 27

written today, before
any retriever saw them:
report only, never tune
15 answerable · 12 not

Every miss gets a reason

Counts first, then the questions behind them.

ANSWER IN THE TOP 3		MRR@10
questions, BM25	29/31	0.85
questions, dense	29/31	0.90
lookups, BM25	30/30	1.00
lookups, dense	19/30	0.56

94% on 31 questions = 79% to 98%: read counts, not decimals

15 MISSES IN THE TOP 3, BY REASON

exact value — dense · lookups — ○○○○○○○○○○○○○ — Day 08 hybrid

vocabulary — BM25 · new dads, wine — ○○ — Day 08 / 10

other section — dense · lost laptop — ○ — Day 11

broader chunk — dense · wine — ○ — Day 09 rerank

the tags say where a fix will be TESTED, not promised

"Will Acme pay for wine with a business dinner?"

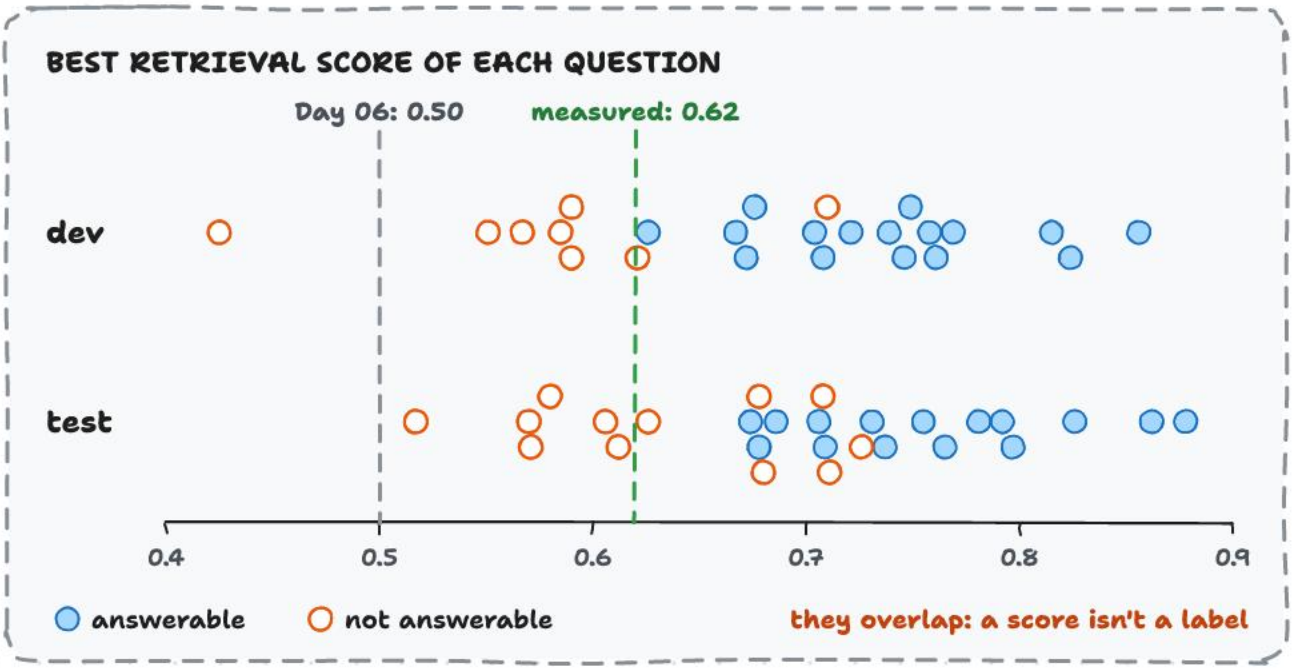
the answer: "Alcohol is not reimbursed."

BM25 ranks it #6 (only "business" matches) · dense #4

a person sees wine = alcohol; neither retriever does

Thresholds, measured

Day 06 chose them by eye. A rule on dev, reported on test.



gate 0.50 (Day 06)

refuses 0 of 12

unanswerable test questions

gate 0.62 (measured)

refuses 6 of 12

and keeps all 15 answerable

rule, written first: highest gate that loses no answerable dev question

THE NAMES RULE: PERFECT ON DEV, WRONG ON TEST

dev: refused Google, France, CEO, Wi-Fi · no false refusals

test: refused 2 answerable questions:

"Tokyo" (the Other cities row applies) · "Should" (2nd sentence)